

Healthcare Appointment No-Show Prediction

Introduction

Hospitals lose time and resources when patients miss appointments. This project predicts no-shows and analyses factors influencing attendance to support better scheduling.

Abstract

The dataset was cleaned, engineered with features such as WaitingDays, AgeGroup and weekdays, analysed, and used to train a Decision Tree classifier. Insights were visualised in Power BI to support operational decisions.

Tools Used

Python, Pandas, Scikit-learn, Matplotlib, Power BI.

Steps Involved

1. Imported and cleaned the dataset.
2. Removed invalid records.
3. Created WaitingDays, AgeGroup and weekday features.
4. Performed exploratory data analysis.
5. Trained a Decision Tree model.
6. Built a Power BI dashboard.

Conclusion

Approximately one in five appointments resulted in a no-show. Waiting time and patient demographics influenced attendance. Predictive analytics can help hospitals identify high-risk patients and optimise scheduling.